Review by: Daniel

Annotation colour legend:
red : general feedback
Pink : minor punctuation/style feedback
Purple : feedback for RQ1
blue : feedback for RQ2
teal : feedback for RQ3
green: feedback for RQ4

Systematic Literature Review Protocol

Applications of Machine and Computer Vision in Software Engineering

Folajinmi Adewole
York University
fj2005@yorku.ca

Yaseen Ejaz Ahmed
York University
yea@yorku.ca

1 Introduction

This protocol defines the methodology for a Systematic Literature Review (SLR) investigating the applications of machine and computer vision in Software Engineering. With the increasing adoption of visual representations, diagrams, screenshots, logs, and interface artifacts throughout the Software Development Life Cycle (SDLC), computer vision techniques are increasingly being explored to automate, support, and enhance software engineering tasks.

The objective of this SLR is to systematically identify and synthesize existing research that applies computer vision and generative AI techniques to software engineering problems. In particular, the review focuses on understanding which software engineering tasks are addressed using computer vision, the types of vision models employed, and the application areas in which generative AI is integrated with vision-based approaches. Additionally, this review examines the challenges encountered when applying computer vision in software engineering contexts and the performance metrics used to evaluate such approaches.

By addressing the defined research questions (RQ1–RQ5), this review aims to provide a structured overview of the current state of research, highlight gaps in existing literature, and inform future research directions at the intersection of computer vision and software engineering.

this feels like a theoretical question that doesn't lead to specific empirical research. What if you asked, for example, "what SE tasks are currently being supported by CV techniques?"

there are many different types of CV models (image classification, object detection, visual question answering, etc.). Which ones are you focusing on? Or are you exploring what types of CV models are used in SE?

this RQ may be out of scope because it asks about genAI + CV, which can be very broad. This can be targeted to SE by asking, for example, "in what ways is genAI+CV used in SE?"

2 Research Questions

The review is guided by the following research questions:

- RQ1: What software engineering tasks are addressed by computer vision?
- RQ2: What computer vision models are employed in research?
- RQ3: What are the primary application areas of generative AI within computer vision?
- RQ4: What challenges are associated with computer vision in software engineering tasks? missing question mark

challenges associated with the use of CV in SE

- RQ5: What are the primary performance metrics used to assess computer vision–based approaches across different software engineering tasks?

3 Datasets

This review considers studies sourced from established digital libraries and indexing platforms commonly used in Software Engineering research. The primary sources include IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Scopus. These repositories provide access to peer-reviewed journal articles and conference proceedings relevant to computer vision and software engineering.

The term *datasets* in this review refers to both:

- Empirical datasets used within the selected studies (e.g., screenshots, UI images, diagrams, logs, bug reports, or visual artifacts), and
- Scholarly datasets in the form of curated research publications retrieved from selected digital libraries.

Both publicly available and proprietary datasets reported in the studies are considered, provided they are used to evaluate computer vision–based approaches in software engineering tasks.

4 Keywords and Search Strings

4.1 Keywords

The following keywords are used to construct the search strategy:

- Computer Vision
- Generative AI
- Deep Learning
- Software Engineering
- Software Development Life Cycle (SDLC)
- Software Development
- Software Testing
- Software Maintenance
- Program Analysis
- Performance Metrics
- User Interface Analysis
- Challenges feels unnecessary

4.2 Search Strings

Search strings are formed by combining computer vision–related terms with software engineering terms. Example search strings include:

You would be able to add more search strings if you had more specific RQs. For example, you could include "vision question answering" and "image understanding" in your set of keywords.

side note: to get the left quotation marks in LaTeX, use backticks: ``quoted text''

- (”computer vision” AND ”software engineering”)
- (”computer vision” AND ”software development”)
- (”computer vision” AND ”software development life cycle”)
- (”computer vision” AND ”software testing”)
- (”computer vision” AND ”software maintenance”)
- (”computer vision” AND ”program analysis”)
- (”computer vision” AND ”user interface analysis”)
- (”deep learning” AND ”computer vision” AND ”software engineering”)
- (”generative AI” AND ”computer vision” AND ”software engineering”)

won't these search strings return subsets of the "CV" + "AI" search string?

- (”computer vision” AND ”software engineering” AND ”performance metrics”)
- (”computer vision” AND ”software engineering” AND challenges) missing quotation marks
- (”computer vision” AND ”software engineering” AND limitations)
- (”generative AI” AND ”software development”)
- (”computer vision” AND ”software engineering”) AND (”performance metrics” OR ”accuracy” OR ”precision” OR ”recall”)

5 Inclusion Criteria

Studies are included in the review if they meet all of the following criteria:

- The study is published in a peer-reviewed journal, conference, or workshop.
- The study was published in the last ten years.
- The study focuses on the application of computer vision techniques within software engineering.
- The research addresses specific SDLC tasks. be consistent with punctuation use
- Provide sufficient methodological detail and empirical evaluation

be consistent about "the study" or "the research" or no prefix at all

6 Exclusion Criteria

Studies are excluded from the review if they meet any of the following conditions:

- Not written in English
- Not aligned with the defined research questions (RQ1–RQ5) this is already covered in the inclusion criteria
- Do not focus on the application of computer vision techniques within software engineering.
- Duplicate or redundant ~~R~~redundant studies that present substantially similar results to the already included work lower case "r" Similar results are okay (we can meta-analyze them); dupliates are removable
- Do not involve computer or machine vision techniques
- Grey literature such as blogs, tutorials, editorials, or non-peer-reviewed reports